

Multi-Head Attention Numerical Example (2 Heads)

1 Setup and Initial Dimensions

We use a sequence length of 3 words, an initial embedding size (d_{model}) of 4, and 2 heads. The attention dimension for each head is $d_k = d_{\text{model}}/2 = 2$.

1.1 Input Embeddings (X)

The 3 words are represented as rows in matrix X :

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

1.2 Head 1 Weight Matrices ($W_Q^{(1)}, W_K^{(1)}, W_V^{(1)}$)

$$W_Q^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad W_K^{(1)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad W_V^{(1)} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$$

1.3 Head 2 Weight Matrices ($W_Q^{(2)}, W_K^{(2)}, W_V^{(2)}$)

$$W_Q^{(2)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}, \quad W_K^{(2)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}, \quad W_V^{(2)} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2 Parallel Head Calculations for Word 1

2.1 Head 1 Execution

Using Word 1 ($x_1 = [1, 0, 1, 0]$), we project to find its query vector:

$$q_1^{(1)} = x_1 \times W_Q^{(1)} = [2, 0]$$

Projecting all inputs yields the key and value spaces for Head 1:

$$K^{(1)} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \\ 1 & 2 \end{bmatrix}, \quad V^{(1)} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \\ 2 & 2 \end{bmatrix}$$

Scores, Scaling ($\sqrt{2} \approx 1.414$), and Softmax:

$$\begin{aligned}\text{Scores} &= \left[q_1^{(1)} \cdot k_1^{(1)}, q_1^{(1)} \cdot k_2^{(1)}, q_1^{(1)} \cdot k_3^{(1)} \right] = [0, 4, 2] \\ \text{Scaled Scores} &= [0, 2.828, 1.414] \\ \text{Softmax Weights} &\approx [0.05, 0.77, 0.18]\end{aligned}$$

Head 1 Output Vector:

$$\text{Output}_1^{(1)} = 0.05 \begin{bmatrix} 2 \\ 1 \end{bmatrix}^T + 0.77 \begin{bmatrix} 0 \\ 2 \end{bmatrix}^T + 0.18 \begin{bmatrix} 2 \\ 2 \end{bmatrix}^T = [0.46, 1.95]$$

2.2 Head 2 Execution

Using Word 1 ($x_1 = [1, 0, 1, 0]$), we find its query vector:

$$q_1^{(2)} = x_1 \times W_Q^{(2)} = [0, 1]$$

Projecting all inputs yields the key and value spaces for Head 2:

$$K^{(2)} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \\ 2 & 1 \end{bmatrix}, \quad V^{(2)} = \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ 2 & 1 \end{bmatrix}$$

Scores, Scaling ($\sqrt{2} \approx 1.414$), and Softmax:

$$\begin{aligned}\text{Scores} &= \left[q_1^{(2)} \cdot k_1^{(2)}, q_1^{(2)} \cdot k_2^{(2)}, q_1^{(2)} \cdot k_3^{(2)} \right] = [1, 2, 1] \\ \text{Scaled Scores} &= [0.707, 1.414, 0.707] \\ \text{Softmax Weights} &\approx [0.24, 0.52, 0.24]\end{aligned}$$

Head 2 Output Vector:

$$\text{Output}_1^{(2)} = 0.24 \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T + 0.52 \begin{bmatrix} 3 \\ 2 \end{bmatrix}^T + 0.24 \begin{bmatrix} 2 \\ 1 \end{bmatrix}^T = [2.04, 1.52]$$

3 Concatenation and Final Output Projection

3.1 Concatenation Step

We combine the vectors from both heads side-by-side to reconstruct our 4-dimensional representation:

$$\text{Concat}_1 = \left[\text{Output}_1^{(1)}, \text{Output}_1^{(2)} \right] = [0.46, 1.95, 2.04, 1.52]$$

3.2 Final Linear Projection (W_O)

We define the linear projection matrix W_O of size 4×4 :

$$W_O = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

The final multi-head attention contextual block for Word 1 is calculated via matrix multiplication:

$$\begin{aligned}
\text{Final Output}_1 &= \text{Concat}_1 \times W_O \\
&= [0.46, 1.95, 2.04, 1.52] \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \\
&= [2.50, 3.99, 3.47, 1.98]
\end{aligned}$$